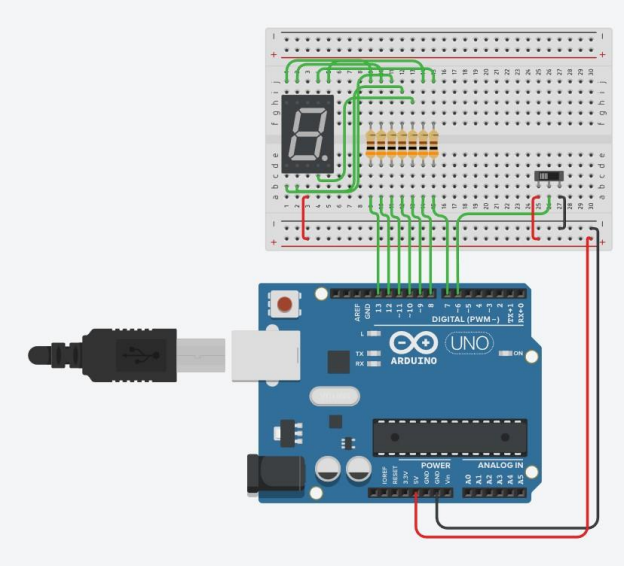
Elita Lestari Setianto (201344008)	**Rangkuman Minggu 11**	Senin, 14 November 2022
		16TNK5043 (Aplikasi Mikrokontroler)
3NK		Ferry Satria, BSEE.,M.T. Rahmawati Hasanah,S.ST.,M.T.

- **Up Down Counter 0-F**

```
int pola[]={0XC0, 0X79, 0X24, 0XB0, 0X99, 0X12, 0X82, 0XF8, 0X80, 0X98, 0X88,
0X83, 0XC6, 0XA1, 0X86, 0X8E };
int i,x;
int pin[7]={7,8,9,10,11,12,13};
int pos[7]={1,2,4,8,0X10,0X20,0X40};
int
mode,j=0;
void
setup()
{
 pinMode(6,INPUT
 );for(i=0; i<7; i++)
   pinMode(pin[i],
OUTPUT);}void display(int
data)
{ x = pola[data];
   for(i=0; i<7;
   i++)
   digitalWrite(pin[i], x &
pos[i]);}void loop()
   { mode =
    digitalRead(6);
    if(mode==HIGH)
    {j++; if(j>15) {j=0;}}
    else
    {j--; if(j<0) {j=15;}
    }
```

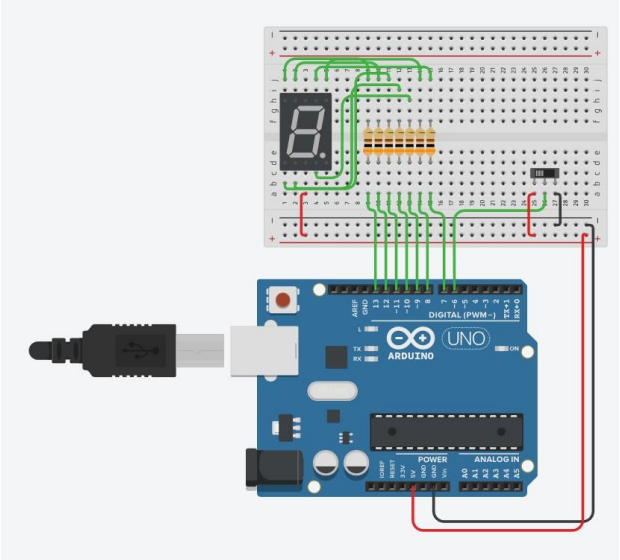
```
  display(j
  );
  delay(50
  0);
 }
```



- **Up Down Counter Mod 10**

```
int pola[]={0XC0, 0X79, 0X24, 0XB0, 0X99, 0X12, 0X82, 0XF8, 0X80, 0X98};
int i,x;
int pin[7]={7,8,9,10,11,12,13};
int pos[7]={1,2,4,8,0X10,0X20,0X40};
int
mode,j=0;
void
setup()
{
 pinMode(6,INPUT
 );for(i=0; i<7; i++)
   pinMode(pin[i],
OUTPUT);}void display(int
data)
{ x = pola[data];
   for(i=0; i<7;
   i++)
   digitalWrite(pin[i], x &
pos[i]);}void loop()
```

```
{ mode =
digitalRead(6);
if(mode==HIGH)
{j++; if(j>15) {j=0;}}
else
{j--; if(j<0) {j=15;}
}
display(j
);
delay(70
0);
}
```



- **DOT MATRIK LAB**

```
.INCLUDE "M8535DEF.INC"

.ORG 0X00

.EQU TIMER_1=0XFF98 RJMP MAIN



.ORG 0X80 DAT:

.DB 0X30,0X36,0X36,0X36,0X6,0
```

```
.DB 0X10,8,4,2,1


MAIN:LDI R16,LOW (RAMEND)

OUT SPL, R16

LDI R16, HIGH (RAMEND) OUT SPH, R16


LDI R16,0XFF OUT DDRA,R16 OUT DDRB,R16


LDI R17,6 LL0:         LDI R18,5

LDI R30,LOW(2*DAT) LDI R31,HIGH(2*DAT)


LL1:    LPM R16,Z

OUT PORTB,R16 ADD R30,R17


LPM R16,Z

OUT PORTA,R16 RCALL D_2MS SUB R30,R17 INC R30

DEC R18 BRNE LL1 RJMP LL0


;----------SUBRUTIN DELAY 2MS-----------

D_2MS: PUSH R16 PUSH R17 LDI R16,1

OUT TIMSK,R16

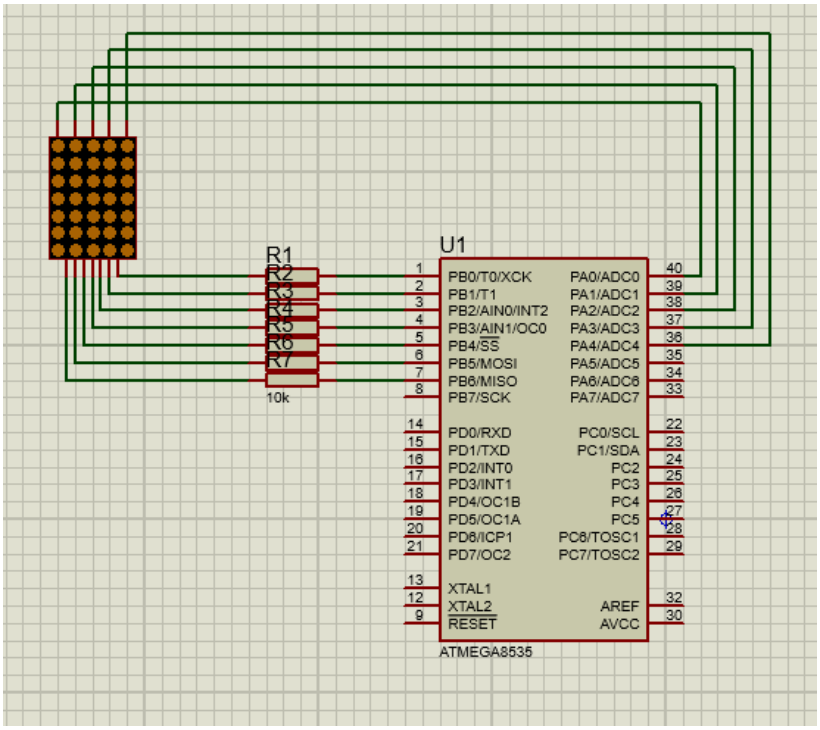
LDI R16, HIGH(TIMER_1) OUT TCNT1H,R16

LDI R16, LOW(TIMER_1) OUT TCNT1L,R16

LDI R16,5
```

```
OUT TCCR1B,R16

LT :        IN R17,TIFR SBRS R17,TOV1
RJMP LT
            LDI R16,4 OUT TIFR,R16
            POP R17
            POP R16
RET
```



- **LCD**

```
#include <LiquidCrystal.h>


const int rs=12,en=11,d4=5, d5=4, d6=3,d7=2;

LiquidCrystal lcd(rs,en,d4,d5,d6,d7); // mesign nilai yg ada
diatasnyaint i,x=0; // inisialisasi nilai i dan x

void setup()

{
```

```
 lcd.begin
 (16,2); //
 lcd.setCu
 rsor(0,0);
 lcd.print("Halo, apa kabar?"); //menampulkan huruf atau
  angkalcd.setCursor(0,1);
  lcd.print("Saya Donny");delay(4000); // mencetak huruf atau angka, ada delaynya

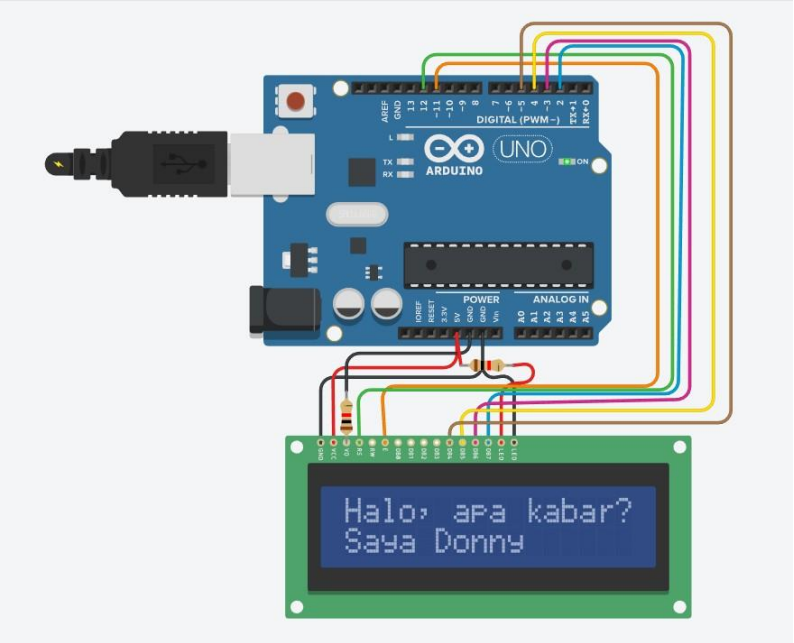

}


void loop()
{ lcd.clear(); //menghapus


 lcd.setCursor(0,0);  // menset ada di baris 1, nanti berpindah kebaris yang
 lainnyafor(i=x; i<=x+7; i++) // nyetak 2 digit 2 digit
  {if(i>99)break; //
   lcd.print(i); // mencetak huruf atau angka
   delay(700);
  }
 lcd.setCursor(0,1);
 for(i=x+8; i<=x+15; i++) // nyetak 1 digit 1 digit
  {if(i>99)break;


   lc
   d.
   p
   ri
   nt
   (i)
   ;
   d
   el
   a
   y(
   7
   0
   0)
   ;
  }
  x=i
 ;
 if(x
 >99
```

```
 )x=
 0;
}
```



Latihan Soal

1. Membuat program menampilkan kata “BISA” bergerak bolak balik (kekiri dan kekanan setiap 5 kali) dari baris 1 ke baris 2 secara berulang. Tentukan sendiri kecepatan pergerakannya

```
#include <LiquidCrystal.h>


const int rs=12,en=11,d4=5, d5=4,
d6=3,d7=2;LiquidCrystal
lcd(rs,en,d4,d5,d6,d7);

int
i,x=0;
void
setup
()

{

 lcd.begin(16,2);

}


void loop() {
lcd.setCursor
(0,0);
lcd.print("BI
SA");
  for (i=0 ; i<=11; i++) {
          lcd.scrollDisplayRight();
          delay(50);

          }

  for (i=0 ; i<=11; i++) {
          lcd.scrollDisplayLe
          ft();delay(50);

          }

  for (i=0 ; i<=11; i++) {
          lcd.scrollDisplayRight();
          delay(50);

          }

  for (i=0 ; i<=11; i++) {
          lcd.scrollDisplayLe
          ft();delay(50);

          }

  for (i=0 ; i<=11; i++) {
          lcd.scrollDisplayRight();
          delay(50);
```

```
                    }
            for (i=0 ; i<=11; i++) {
                    lcd.scrollDisplayLe
                    ft();delay(50);
                    }
            for (i=0 ; i<=11; i++) {
                    lcd.scrollDisplayRight();
                    delay(50);
                    }
            for (i=0 ; i<=11; i++) {
                    lcd.scrollDisplayLe
                    ft();delay(50);
                    }
for (i=0 ; i<=11; i++) { lcd.scrollDisplayRight(); delay(50);

}

for (i=0 ; i<=11; i++) { lcd.scrollDisplayLeft(); delay(50);

}

lcd.clear(); lcd.setCursor(0,1); lcd.print("BISA");

for (i=0 ; i<=11; i++) { lcd.scrollDisplayRight(); delay(50);

}

for (i=0 ; i<=11; i++) { lcd.scrollDisplayLeft(); delay(50);

}

for (i=0 ; i<=11; i++) { lcd.scrollDisplayRight(); delay(50);

}

for (i=0 ; i<=11; i++) { lcd.scrollDisplayLeft(); delay(50);

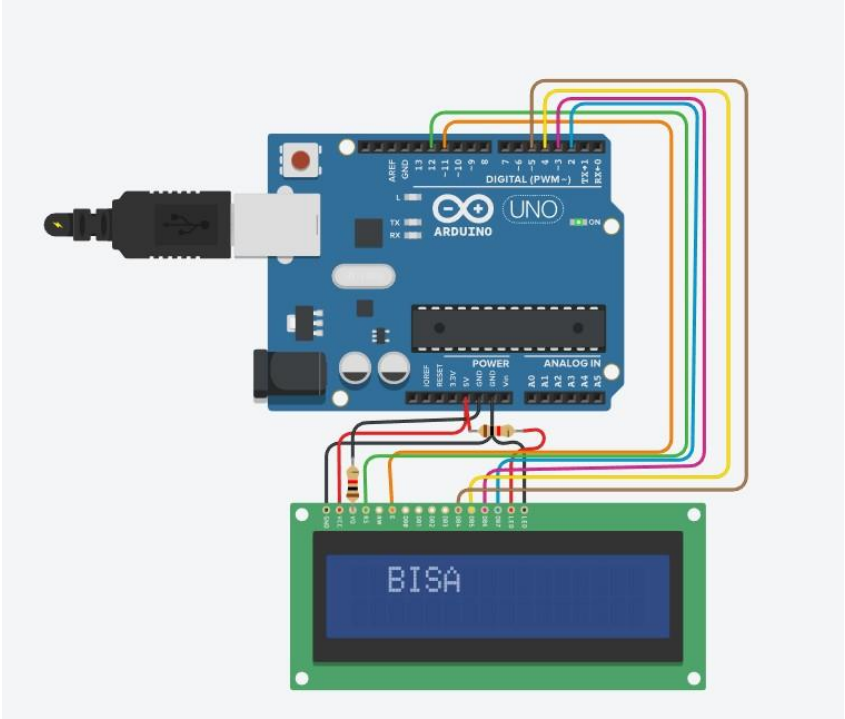
}

for (i=0 ; i<=11; i++) { lcd.scrollDisplayRight(); delay(50);

}

for (i=0 ; i<=11; i++) { lcd.scrollDisplayLeft(); delay(50);
```

```
}
for (i=0 ; i<=11; i++) { lcd.scrollDisplayRight(); delay(50);
}
for (i=0 ; i<=11; i++) { lcd.scrollDisplayLeft(); delay(50);
}
for (i=0 ; i<=11; i++) { lcd.scrollDisplayRight(); delay(50);
}
for (i=0 ; i<=11; i++) { lcd.scrollDisplayLeft(); delay(50);
                }
        lcd.clear();
        }
```



2. Membuat 4 counter (menghitung naik secara berulang), Tentukan sendiri kecepatan pergerakannya

   - Ujung kiri atas up counter desimal mod 10
   - Ujung kanan atas up counter desimal mod 100
   - Ujung kiri bawah down counter HD mod 16
   - Ujung kanan bawah down counter HD mod 256 (FF-00)

3. Membuat program seperti soal no 1, selain bergerak tapi berkedip juga

```
#include <LiquidCrystal.h>


const int rs=12,en=11,d4=5, d5=4,
d6=3,d7=2;LiquidCrystal
lcd(rs,en,d4,d5,d6,d7);

int i,x=0;

void setup()
  {
  lcd.begin(
  16, 2);
}



void loop() { lcd.setCursor(0,0); lcd.print("BISA");



for (int positionCounter = 0; positionCounter < 12; positionCounter++) {
lcd.scrollDisplayRight();

delay(50); lcd.noDisplay(); delay(50); lcd.display(); delay(50);

}



for (int positionCounter = 0; positionCounter < 12; positionCounter++) {
lcd.scrollDisplayLeft();
```

```
delay(50); lcd.noDisplay(); delay(50); lcd.display(); delay(50);

}


for (int positionCounter = 0; positionCounter < 12; positionCounter++) {
lcd.scrollDisplayRight();

delay(50); lcd.noDisplay(); delay(50); lcd.display(); delay(50);

}


for (int positionCounter = 0; positionCounter < 12; positionCounter++) {
lcd.scrollDisplayLeft();

delay(50); lcd.noDisplay(); delay(50); lcd.display(); delay(50);

}


for (int positionCounter = 0; positionCounter < 12; positionCounter++) {
lcd.scrollDisplayRight();

delay(50); lcd.noDisplay(); delay(50); lcd.display(); delay(50);

}

lcd.clear();


lcd.setCursor(0,1); lcd.print("BISA");


for (int positionCounter = 0; positionCounter < 12; positionCounter++) {
lcd.scrollDisplayRight();

delay(50); lcd.noDisplay(); delay(50); lcd.display(); delay(50);

}
```

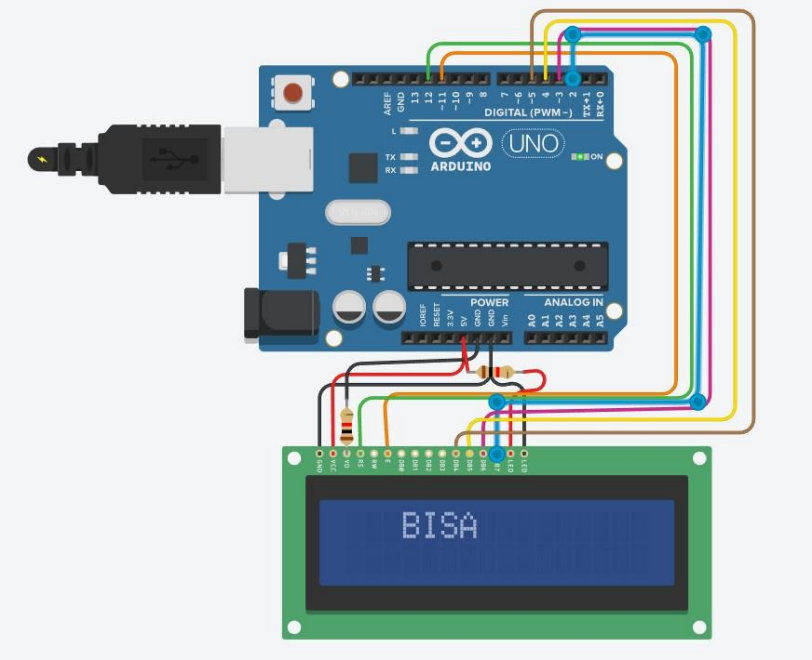
```
for (int positionCounter = 0; positionCounter < 12; positionCounter++) {
lcd.scrollDisplayLeft();
delay(50); lcd.noDisplay(); delay(50); lcd.display(); delay(50);
}

for (int positionCounter = 0; positionCounter < 12; positionCounter++) {
lcd.scrollDisplayRight();
delay(50); lcd.noDisplay(); delay(50); lcd.display(); delay(50);
}

for (int positionCounter = 0; positionCounter < 12; positionCounter++) {
lcd.scrollDisplayLeft();
delay(50); lcd.noDisplay(); delay(50); lcd.display(); delay(50);
}

for (int positionCounter = 0; positionCounter < 12; positionCounter++) {
lcd.scrollDisplayRight();
delay(50); lcd.noDisplay(); delay(50); lcd.display(); delay(50);
}

        lcd.cl
        ear();
        delay
        (150)
        ;
```



4. Membuat program UP/Down counter desimal mod 100 (00-99) yang mode up/downnya dikendalikan secara polling (pilih sendiri) pin untuk menghubungkan push button pollingnya.